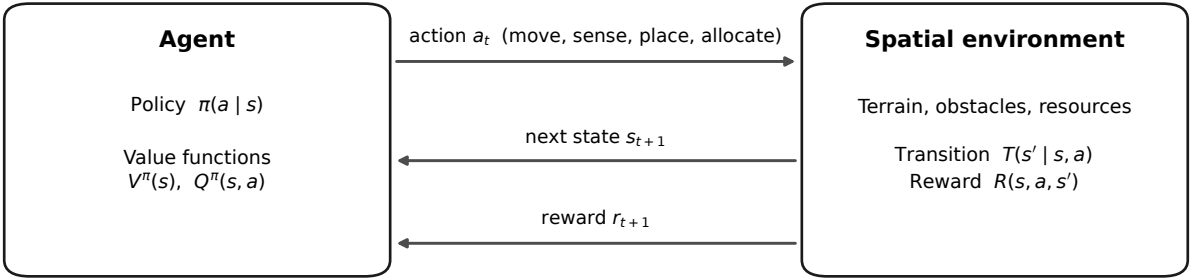
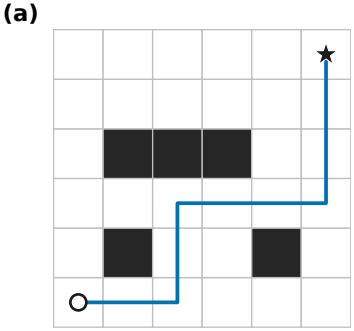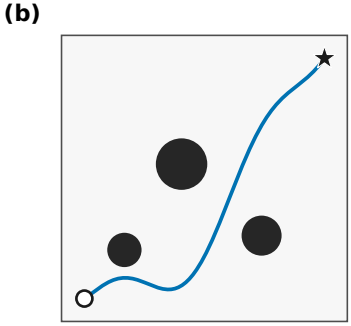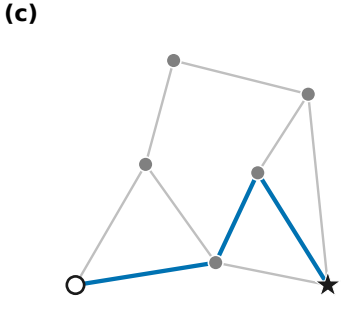
$$V^{\pi}(s) = \mathbb{E}_{\pi}\Big[\sum_{t\geq 0}\gamma^{t} r_{t+1} \mid s_0 = s\Big] = \sum_{a}\pi(a \mid s)\sum_{s'}T(s' \mid s,a)[R(s,a,s') + \gamma V^{\pi}(s')]$$